Econ 131
Spring 2026
Danny Yagan

Problem Set 2

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- Write or type your answers clearly and in dark ink (physical or electronic ink) so that your responses are legible.
- Tag each of your answers on Gradescope so that it is clear what responses are to which questions.
- **Although you may work in groups**, each student must submit individual sets of solutions. You must note the names of other students that you worked with. Write their names here:

1. True/False/Uncertain with Explanation [25 points]

Determine whether each statement is true, false, or uncertain and explain why. Answers with no explanation will receive no points. [5 points each]

(a) A dividend income tax cannot raise revenue without reducing investment.

FALSE. A dividend income tax levied on net profits (revenues minus capital cost) does not distort investment. In this case, the firm's profit-maximization FOC is $(1-\tau)[F'(K)-r]=0$, which gives $F'(K)=r$, identical to the no-tax benchmark. The tax only takes a fraction of economics rents, leaving the marginal incentive to invest unchanged. Since the tax still raises positive revenue whenever $F(K)-rK>0$, revenue is raised without reducing investment.

Lecture 9 : Business taxation. taxes on rents vs. taxes on gross profit

(b) If the income distribution in the top tax bracket becomes thinner (individuals above the bracket threshold become more concentrated at incomes near the threshold), the government should increase the top marginal tax rate.

False. The Saez optimal top tax rate formula is "$\tau^* = \frac{1}{1+a \cdot e}$", where "a" is the Pareto parameter and "e" is the ETI. A thinner distribution, with top earners concentrated near the bracket threshold, means the average income above the threshold "zm" is close to "z^*", making the dominator "$(zm - z^*)$" small and therefore a large. A higher "a" raises the dominator "$(1 + a \cdot e)$", which lowers the "τ^*". Intuitively, when high earners are concentrated just above the threshold, they are especially responsive to tax increases, so the revenue-maximizing rate is lower.

Lecture 7: Optimal top tax rate, Saez formula, Perato parameter

(c) If leisure is a normal good, then an increase in non-labor income creates a negative income effect on labor supply.

True. An increase in non-labor income R is a pure income effect. it shifts the budget constraint outward without changing the net-of-tax wage, so there is no substitution effect. Since leisure is a normal good, the income effect causes the individual to demand more leisure. More leisure means fewer hours of work, so labor supply falls. The income effect on labor supply is therefore negative, which confirms the statement.

Labor supply theory: income effects, normal goods, non-labor income.

(d) A lump-sum tax in period 1 that is independent of savings changes the intertemporal relative price between current and future consumption.

False. A lump-sum tax in period 1 reduces disposable income from y_1 to $(y_1 - T)$ but does not affect the return on savings r. The intertemporal budget constraint becomes:

$$c_1 + \frac{c_2}{1+r} = (y_1 - T) + \frac{y_2}{1+r}$$

which is a parallel inward shift. The slope $(1+r)$ is unchanged. Since the relative price of future vs. current consumption is determined by the slope $(1+r)$, a lump-sum tax creates only an income effect, not a substitution effect. By contrast, a capital income tax changes the net return to $r(1-t)$, rotating the constraint and altering relative prices. The lump-sum tax leaves the intertemporal relative price entirely unchanged.

Lecture 10: Intertemporal budget constraint, lump-sum vs. capital income taxes.

(e) The Pareto parameter for the entire income distribution is 1/2.

False. The Pareto parameter "a" must be strictly greater than 1 for the income distribution to have a finite mean. This follows from the Pareto density: $E[z]$ converges only when $a > 1$. A value of $a = 1/2$ would imply an infinite average income, which is empirically impossible. Additionally, the formula relating the average income above the threshold to the threshold itself is $\frac{z_m}{z^*} = \frac{a}{a-1}$, which is only meaningful when $a > 1$. Empirically, the Pareto parameter for the upper tail of the U.S. income distribution is approximately $1.5 \sim 2.5$.

Lecture 7: Pareto parameter, properties of income distributions.

2. Difference-in-differences [15 points] Lecture 8B (Event Studies and DiD)

Recall the graph from Kleven-Landais-Saez (American Economic Review 2013) that we drew in class of the number of Sports and Entertainment foreign workers in Denmark. Recall the regression that we ran to compute the DD estimate of the impact of the 1991 tax reform on the number of top-earning Sports and Entertainment foreign workers in Denmark on average 1991-2005. Suppose that a policymaker asks for the impact of the 1991 tax reform on the number of top-earning Sports and Entertainment foreign workers in Denmark in 2005.

(a) [5 points] Is the 2005 impact bigger or smaller than the average 1991-2005 impact?

The 2005 impact is bigger than the average 1991-2005 impact. In the KLS event study graph, the number of top-earning foreign S&E workers in Denmark increased progressively after the 1991 tax reform. The treatment effect grew over-time as more foreign workers relocated in response to the preferential tax rate. Because the effect was smaller in the early post-reform years and larger by 2005, the 2005 specific treatment effect lies above the average effect computed over the entire 1991-2005 period.

(b) [10 points] How would you compute the 2005 impact in a single regression? A complete answer should include the regression equation that you would estimate and the data sample you would use. The regression equation should be written in math like in the Lecture 8B slides, not in Stata or other computer code. (Note: There is more than one way to do this regression, but there is a single simple way based on what we have already done in class.)

To compute the 2005 specific impact, I would use the same regression structure as the class DiD, but restrict the sample to pre-reform observations and year 2005 only. The regression is:

$$Y_{it} = \alpha + \beta \cdot Treated_i + \gamma \cdot Post2005_t + \delta \cdot (Treated_i \cdot Post2005_t) + \varepsilon_{it}$$

where Y_{it} is the number of top-earning foreign workers in sector i at time t, $Treated_i = 1$ for S&E workers and 0 for the control group of other top earners, and $Post2005_t = 1$ if $t = 2005$ and 0 if t is in the pre-reform period. The coefficient δ identifies the causal 2005 treatment effect by comparing the pre-to-2005 change in the S&E group to the pre-to-2005 change in the control group. The key modification from the class regression is that by dropping the intermediate post-reform years from the sample, the interaction term isolates the 2005 effect rather than averaging over 1991–2005.

3. Business Taxation and Investment [26 points] Lecture 9 (Business Taxation)

Consider a firm that chooses capital K to maximize profit. Let $F(K)$ denote the firm's gross profit from using capital, where $F'(K) > 0$ and $F''(K) < 0$. Let r denote the per-unit cost of capital.

Throughout, assume the firm chooses K optimally and interior solutions exist.

(a) [4 points] Suppose there is no tax. The firm solves

$$\max_K \; F(K) - rK.$$

Write the first-order condition for the firm's optimal choice of capital.

$$\max_K F(K) - rK \Rightarrow \frac{d}{dK}[F(K) - rK] = 0$$

$$\Rightarrow F'(K) - r = 0$$

Therefore : FOC : $F'(K) = r$

(b) [5 points] Now suppose the government imposes a **business income tax** at rate τ on **gross profit**, so the firm solves

$$\max_K \; (1-\tau)F(K) - rK.$$

Derive the first-order condition. Does this tax increase, decrease, or leave unchanged the firm's desired level of investment? Briefly explain.

$$\max_K (1-\tau)F(K) - rK \Rightarrow \frac{d}{dK}[(1-\tau)F(K) - rK] = 0$$

$$\Rightarrow (1-\tau)F'(K) - r = 0 \Rightarrow \text{FOC}: \quad F'(K) = \frac{r}{1-\tau}$$

Since $F''(K) < 0$, diminishing marginal product, a higher required $F'(K)$ means the firm must be at a lower K to satisfy the FOC. Therefore investment decreases.

(c) [5 points] Instead suppose the business income tax applies to profits **after deducting capital costs**, so the firm solves

$$\max_K \, (1-\tau)\big[F(K) - rK\big].$$

Derive the first-order condition. Does this tax affect the firm's desired investment? Briefly explain.

$\max_K (1-\tau)[F(K)-rK] \Rightarrow \frac{d}{dK}\left[(1-\tau)[F(K)-rK]\right] = 0$

$\Rightarrow (1-\tau)[F'(K)-r] = 0 \Rightarrow$ since $(1-\tau) \neq 0$

$\Rightarrow$ FOC : $F'(K) = r$

The result is identical to the FOC in part (a). The optimal K is exactly the same as in the no-tax case. Investment is unchanged.

(d) [4 points] In 2–3 sentences, explain the economic intuition for why the answers in parts (b) and (c) are different.

The key difference is whether capital costs are deductible from the tax base. In part (b), the firm pays the full rental cost r but only keeps $(1-\tau)$ of gross revenue, creating a tax wedge that raises the effective cost of capital above r and reduces investment. In part (c), capital costs are fully deductible, so the government proportionally shares both the returns and the costs of investment. This symmetry means the tax does not alter the marginal cost-benefit comparison the firm faces, leaving the optimal K unchanged. Deductibility of capital costs is what determines whether a business income tax distorts investment.

(e) [4 points] Historically, many capital costs have not been deductible immediately from taxable profits. Explain why this makes the tax system less like the case in part (c) and more like the case in part (b).

When capital costs are deducted gradually over time rather than immediately, the present value of the tax deductions is less than the actual cost of the investment, since future deductions are discounted. This means the firm cannot fully offset its capital expenditure against current taxable income, so the tax effectively applies to more than just net profit. In the extrem case where no deduction is ever allowed, the system is exactly like part (b), the firm bears the full cost of capital but only keeps $(1-\tau)$ of the gross return. The slower the depreciation schedule, the lower the present value of deductions and the more closely the system resembles the investment-distorting case in (b) rather than the natural case in (C).

(f) [4 points] Suppose the tax system is the one from part (b). Now suppose that the government offers an investment tax credit c: for every dollar spent on capital, the firm gets to reduce its tax bill by \$$c$. What value does c need to be in order for the new tax system as a whole to not distort the firm's choice of capital relative to (a)?

Setup: ① Total dollars spent on capital $= rK$

② Tax credit $= c \times rK = crK$

$$\Rightarrow \max_K (1-\tau)F(K) - rK + crK = (1-\tau)F(K) - (1-c)rK$$

$$\Rightarrow \frac{d}{dK}\left[(1-\tau)F(K) - (1-c)rK\right] = 0$$

$$\Rightarrow (1-\tau)F'(K) = (1-c)r \Rightarrow F'(K) = \frac{(1-c)r}{(1-\tau)}$$

Set equal to the no-tax FOC from (a):

$$\frac{(1-c)r}{(1-\tau)} = r$$

$\Rightarrow 1-c = 1-\tau$ Therefore $c = \tau$

Lecture 8 (Labor Supply and Taxable Income)

4. Labor Income Taxes and Labor Supply [18 points]

An individual chooses labor supply $n \geq 0$. Consumption is

$$c = (1-\tau)wn + R,$$

where w is the pre-tax wage, τ is the labor income tax rate, and R is non-labor income.

Consider the two utility functions

$$\text{(I)} \quad u_1(c,n) = c - \frac{n^{1+1/\varepsilon}}{1+1/\varepsilon},$$

$$\text{(II)} \quad u_2(c,n) = \log(c) - \alpha\frac{n^{1+1/\varepsilon}}{1+1/\varepsilon},$$

where $\varepsilon > 0$ and $\alpha > 0$.

(a) [3 points] For case (I), derive the first-order condition for optimal labor supply. Show that labor supply depends on the net-of-tax wage $(1-\tau)w$.

Setup: ① $c = (1-\tau)wn + R$

② $u_1 = c - \frac{n^{1+1/\varepsilon}}{1+\frac{1}{\varepsilon}}$

$\Rightarrow u_1 = (1-\tau)wn + R - \frac{n^{1+1/\varepsilon}}{1+1/\varepsilon}$

$\Rightarrow \frac{\partial u_1}{\partial n} = (1-\tau)w - n^{1/\varepsilon} = 0$

$\Rightarrow n^{1/\varepsilon} = (1-\tau)w$

$\Rightarrow n^* = [(1-\tau)w]^{\varepsilon}$

(b) [3 points] In case (I), if the tax rate τ rises, is there an income effect on labor supply? Is there a substitution effect? What happens to labor supply overall?

In case (I), there is no income effect. Because u_1 is quasi-linear, the marginal utility of consumption is constant and does not depend on income. Changes in the tax rate that affect the individual's income have no effect on labor supply through this channel. There is a substitution effect: when τ rises, the net-of-tax wage $(1-\tau)w$ falls, making each hour of work less rewarding and leisure relatively cheaper, so the individual work less. Overall, labor supply falls when τ rises, unambiguously driven by the substitution effect alone: $n^* = [(1-\tau)w]^{\varepsilon}$ decreases as $(1-\tau)$ decreases.

(c) [3 points] In case (II), if the tax rate τ rises, explain why labor supply is affected by both an income effect and a substitution effect. Do these two effects go in the same direction or opposite directions?

In case (II), $\log(C)$ is concave in C, so the marginal utility of consumption $1/C$ depends on the individual's income level. When τ rises, two effects occur simultaneously. Substitution effect: the net-of-tax wage $(1-\tau)w$ falls, making leisure cheaper relative to consumption, so the individual substitutes toward leisure and work less. Income effect: higher τ reduces the individual's income. Since leisure is a normal good, being poorer leads to less demand for leisure, so the individual works more. These two effects go in opposite directions, making the net effect on labor supply ambiguous under case (II).

(d) [3 points] Which case would generate a larger gap between the compensated and uncompensated labor-supply elasticities? Explain briefly.

Case (II) generates a large gap between compensated and uncompensated labor-supply elasticities. By the Slutsky decomposition, the difference between the two elasticities equals the income effect. In case (I), utility is quasi-linear in c, so the marginal utility of compostion is constant and there is no income effect, the compansated and uncompensated elasticites are identical. In case (II), log utility creates a genuine income effect, so the uncompensated elasticity is smaller than the compensated elasticity, producing a positive gap.

(e) [3 points] Suppose the government increases τ but also adjusts R so that the individual can still reach the same utility level as before. Is this change closer to an uncompensated or a compensated tax change? In this situation, which effect on labor supply is being isolated?

This is a compensated tax change. By adjusting R so that the individual remains on the same indifference curve, the government neutralizes the income effect that would otherwise accompany the tax increase. The individual is fully compensated for the income loss, so they do not need to work more to maintain their standard of living. The only behavioral response is the substitution effect: the net-of-tax $(1-\tau)w$ has fallen, making leisure cheaper, so the individual works less. This experiment isolates the substitution effect and measures the compensated labor-supply elasticity.

(f) [3 points] In empirical public finance, labor-supply responses are often summarized using the elasticity of labor income with respect to the net-of-tax rate, $1-\tau$. Based on your answers above, would you expect this elasticity to be larger in absolute value under case (I) or case (II)? Explain briefly.

The elasticity of labor income with respect to the net-of-tax rate $(1-\tau)$ is larger in case (I). In case (I), there is no income effect, so the full elasticity equals ε, driven entirely by the substitution effect. Formally, $n^* = [(1-\tau)w]^{\varepsilon}$, so $(1-\tau)$ is exactly ε. In case (II), the income and substitution effects work in opposite directions: when $(1-\tau)$ rises (tax cut), the substitution effect increases labor supply, but the income effect reduces it. Individual is richer, buys more leisure. These partially cancel, so the uncompensated elasticity in case (II) is smaller in absolute value than in case (I). Since the empirical ETI captures the uncompensated total behavioral response, we would expect a larger elasticity under case (I).